

## EXPERIMENT 6

### DATA VISUALIZATION USING POWER BI

#### Aim

To import, transform, analyze, and visualize data using Power BI Desktop and create interactive dashboards and reports for decision-making.

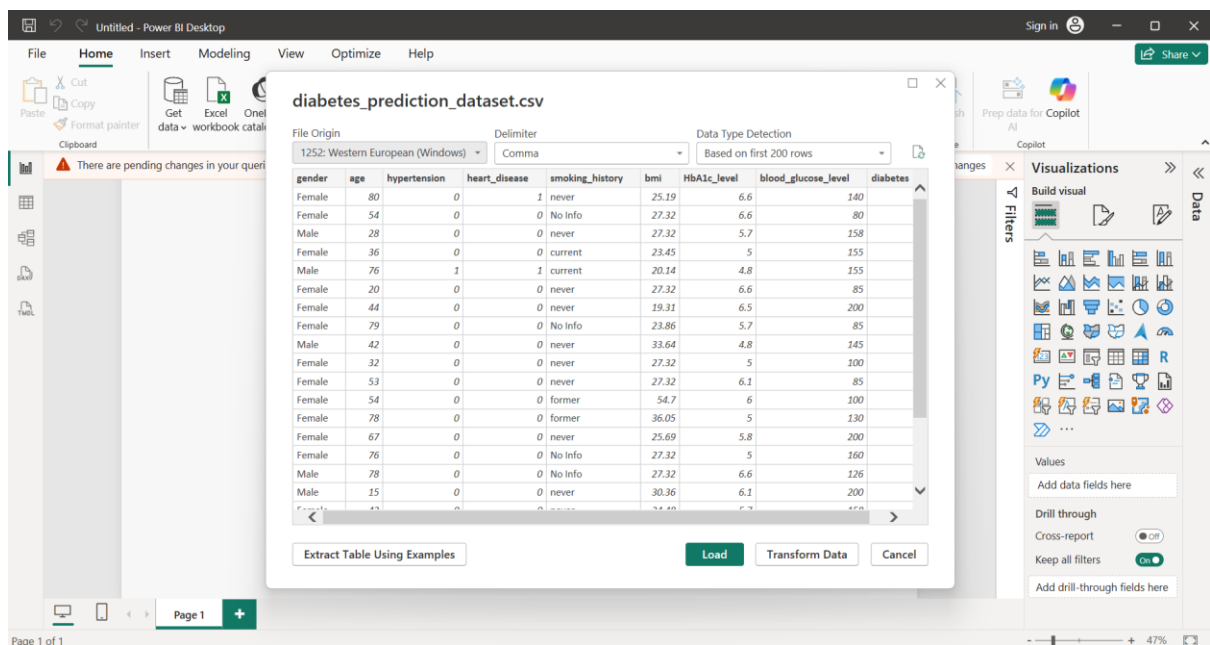
#### Algorithm 1 Importing Data into Power BI

Step 1 Open Power BI Desktop

Step 2 Select: Home → Get Data

Step 3 Choose CSV

Step 4 Browse dataset and select it



#### Algorithm 2 Data Transformation using Power Query

Step 1 Select Transform Data

Step 2 Open Power Query Editor

Step 3 Remove duplicates

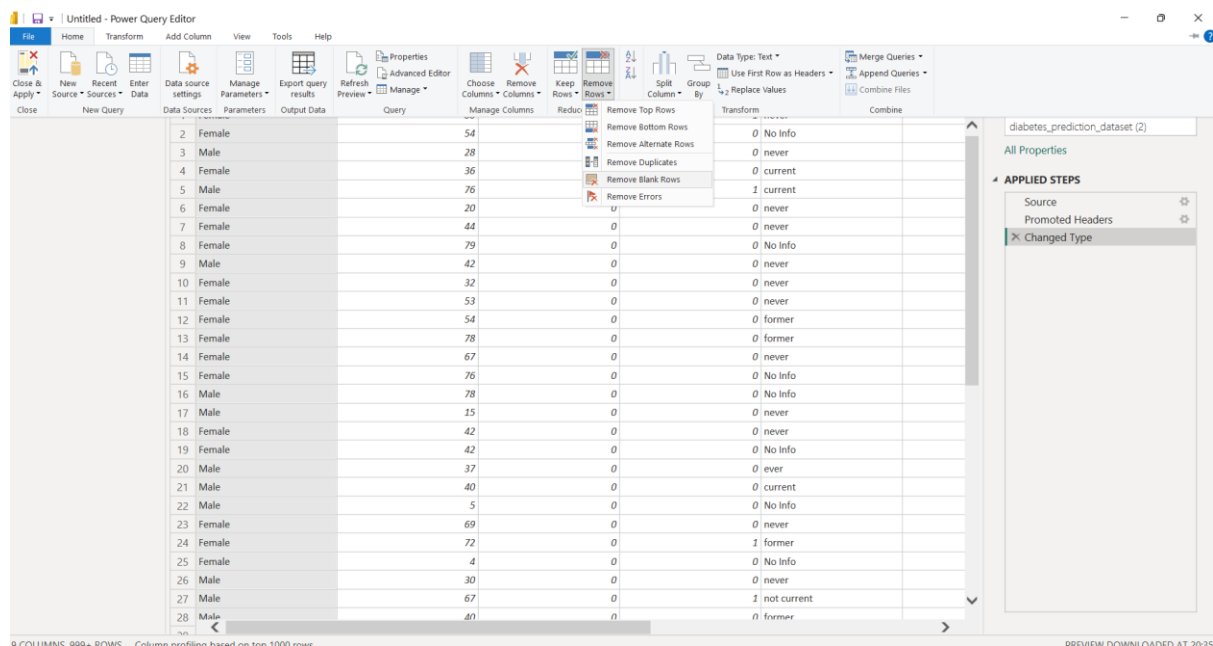
Step 4 Handle null values

Step 5 Rename columns

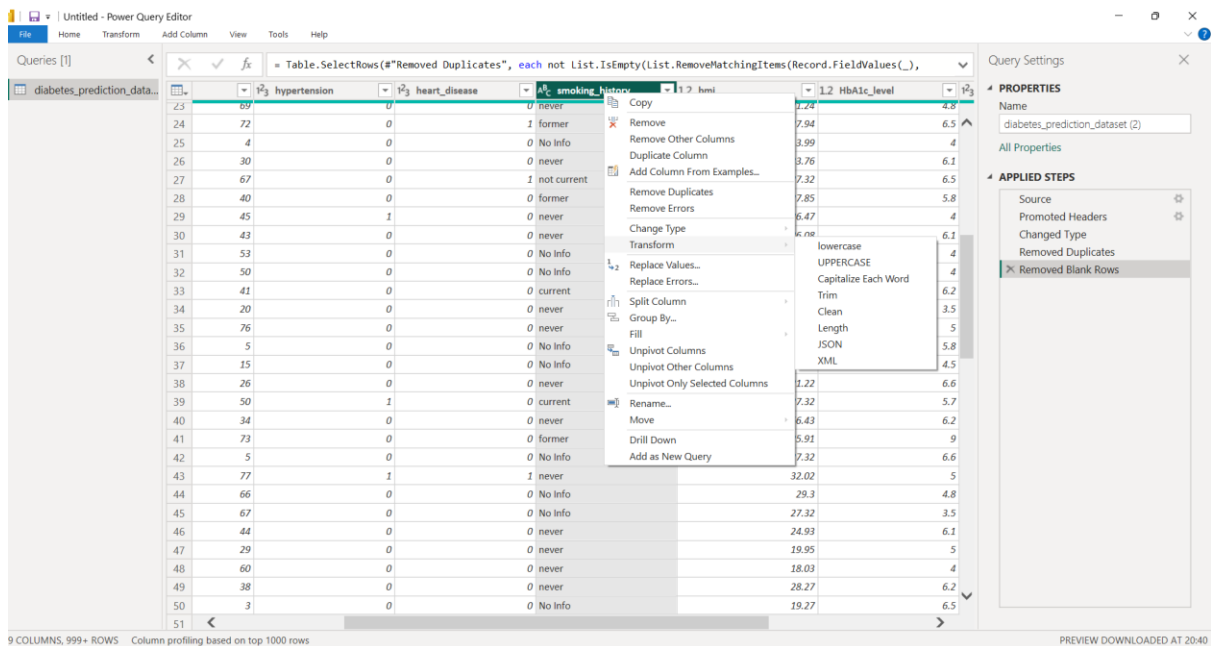
Step 6 Apply changes

## Procedure

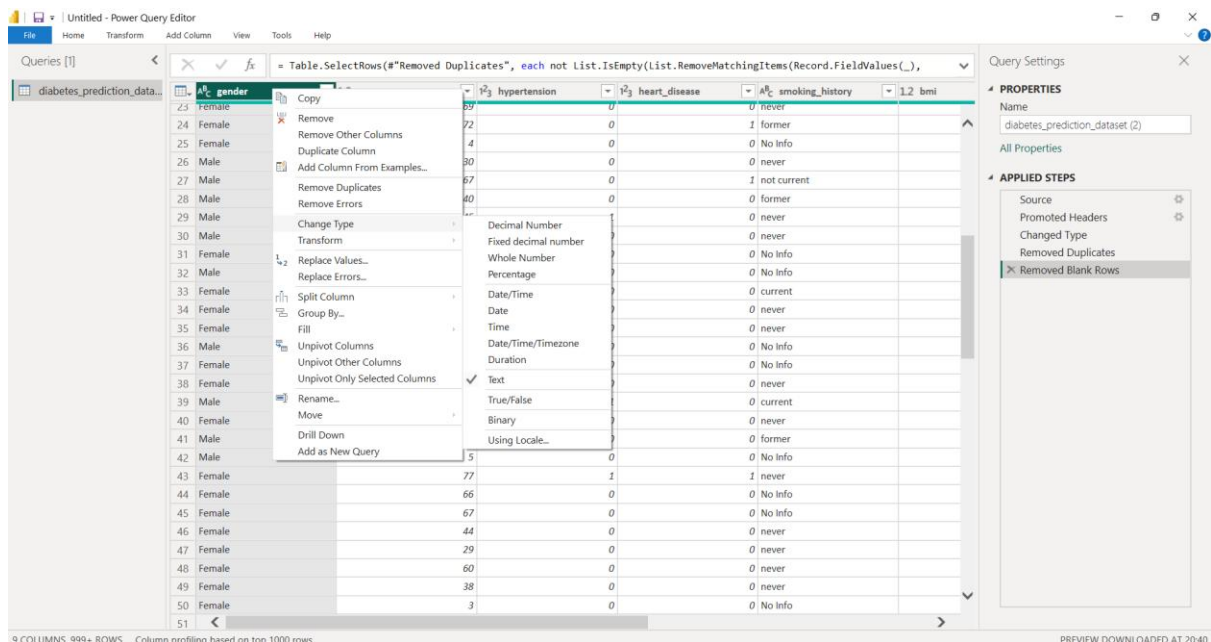
1. The data preview window will appear. Select Transform Data instead of Load to open the Power Query Editor.
2. In the Power Query Editor, examine the imported table and verify that all required columns are present, such as OrderID, OrderDate, CustomerName, Region, Quantity, UnitPrice, and Discount.
3. Remove blank rows from the dataset by selecting Home → Remove Rows → Remove Blank Rows. This removes completely empty records that could interfere with analysis and calculations.



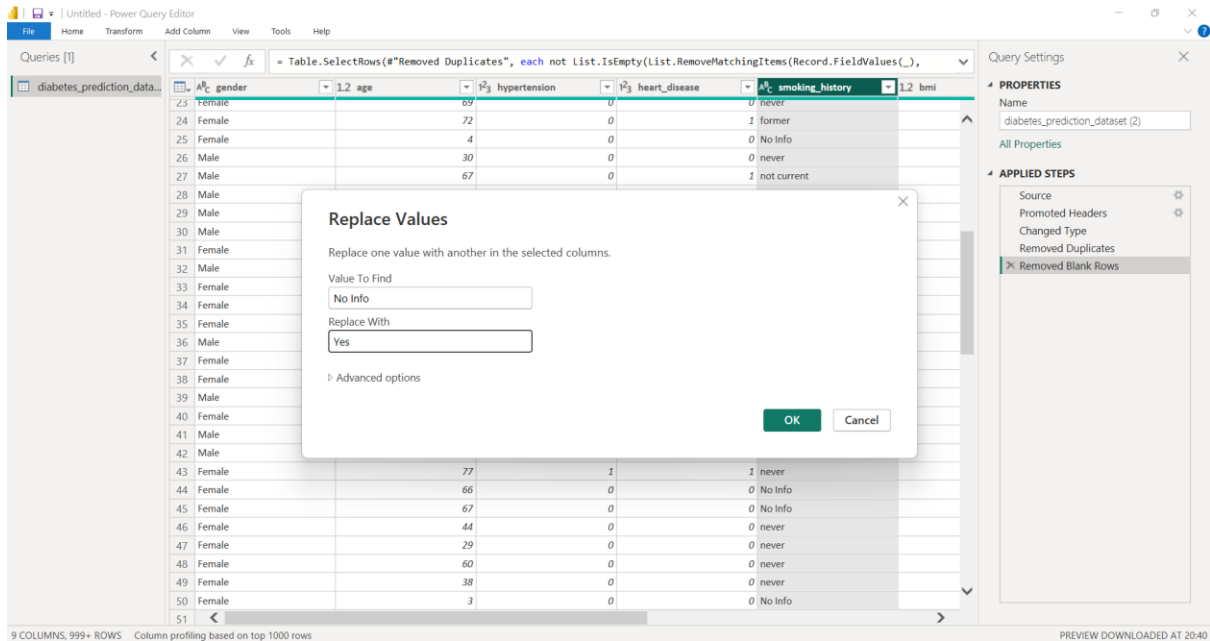
4. Remove duplicate records by selecting all columns in the table using Ctrl+A on the column headers and choosing Home → Remove Rows → Remove Duplicates. Power Query will remove records that contain identical values across all selected columns.
5. Standardize the values in the gender column. Right-click the gender column, select Transform, and apply appropriate text transformation if required. This ensures that values such as female, Female, and FEMALE are represented consistently.



6. Standardize the gender column by selecting it and choosing Transform → Format → Trim to remove any unnecessary spaces from the gender values. Then select Transform → Format → Capitalize Each Word if required so that gender values are consistently represented as “Male” and “Female”. This ensures that inconsistent text formatting does not create separate categories during visualization.

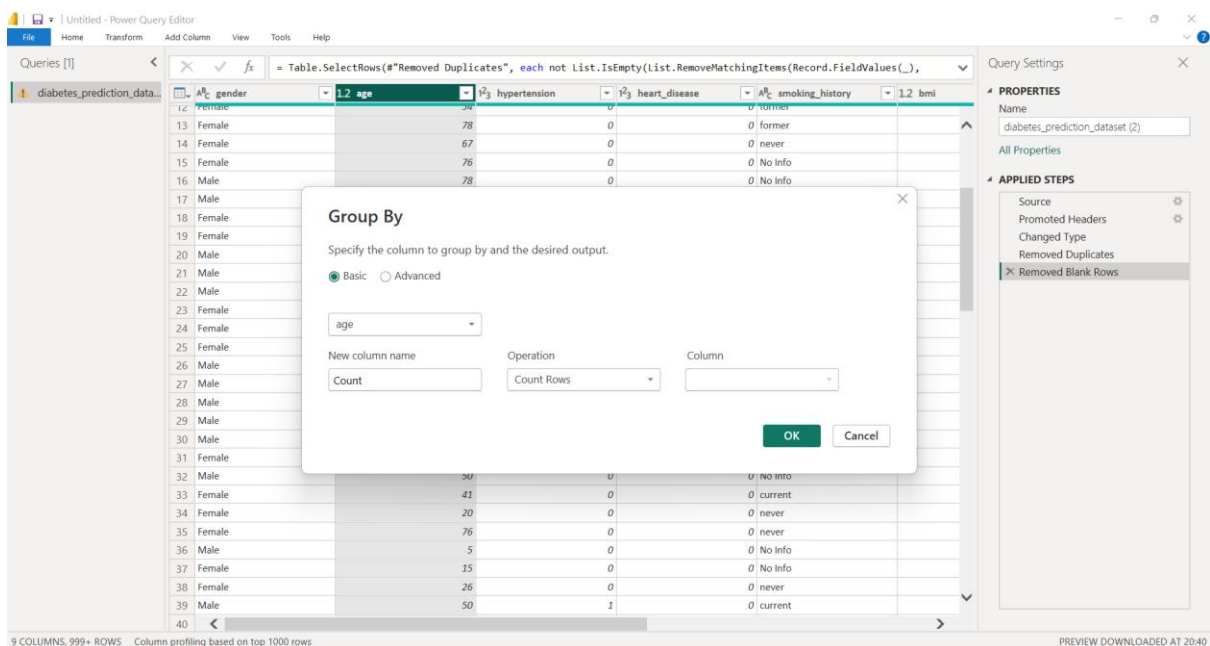


7. Handle missing values in the smoking\_history column. Right-click smoking\_history, select Replace Values, and replace blank or null values with “No Info”. This provides a consistent category for patients whose smoking history is not available.



8. Handle missing values in the numerical columns. Check the bmi, HbA1c\_level, and blood\_glucose\_level columns for null values. If null values are present, replace them with an appropriate statistical value such as the median of the respective column. This prevents missing numerical values from affecting calculations and visualizations.

9. Identify and remove invalid age and health values. Select the age column and apply a filter using the arrow next to the column name to exclude records where age is less than 0. Similarly, verify that bmi, HbA1c\_level, and blood\_glucose\_level contain valid positive values. Negative or impossible numerical values are treated as invalid data because these measurements cannot normally contain negative values.



10. Create a derived column for Age Group. Select Add Column → Custom Column and name the new column Age Group. Enter the following formula:

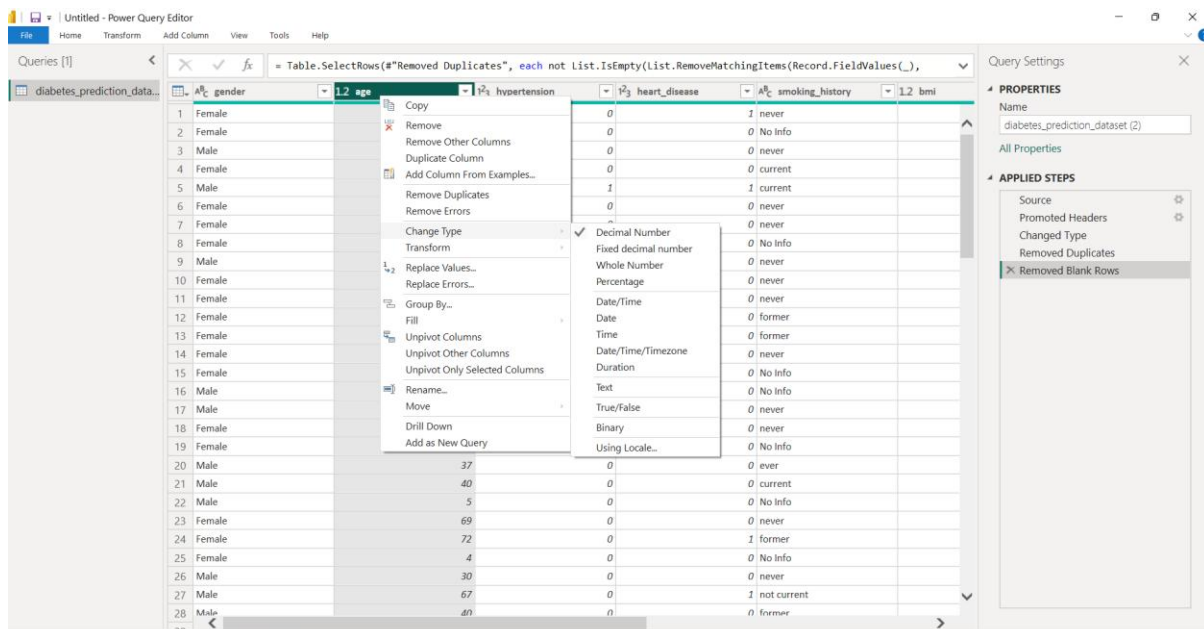
if [age] < 30 then "Young"

else if [age] < 50 then "Middle Age"

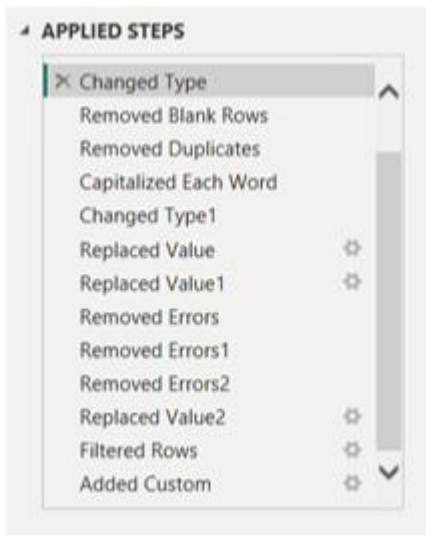
else "Senior"

This categorizes patients into three age groups based on their age and provides a useful categorical variable for analysis and visualization.

11. Verify and correct the data types of all columns. Set age to Whole Number, hypertension to Whole Number, heart\_disease to Whole Number, bmi to Decimal Number, HbA1c\_level to Decimal Number, blood\_glucose\_level to Whole Number, and diabetes to Whole Number. Set gender, smoking\_history, and Age Group to Text. Correct data types ensure that numerical calculations and categorical visualizations work correctly.



12. Review the Applied Steps pane on the right side of the Power Query Editor. Confirm that all transformation steps have been recorded in the correct order. If an error is present, select the corresponding step and correct the transformation. Verify that gender values have been standardized, missing smoking-history values have been handled, invalid numerical values have been removed, and the Age Group column has been created correctly.

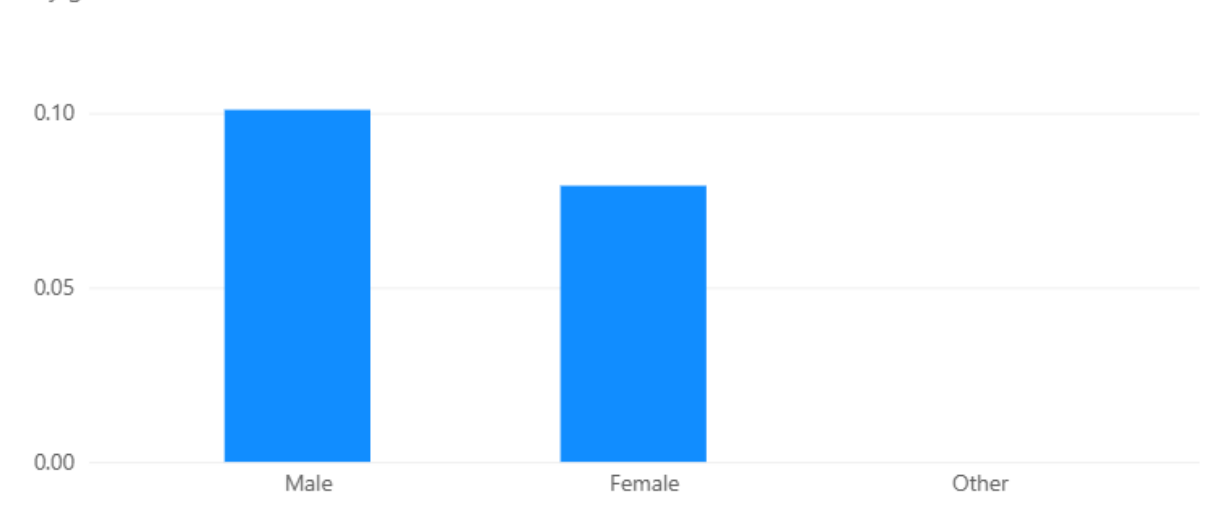


13. After completing all cleaning and transformation operations, select Home → Close & Apply. Power BI will apply the transformations and load the cleaned diabetes dataset into the data model.

14. Switch to the Data/Table view and inspect the transformed table. Verify that duplicate rows, blank rows, inconsistent gender values, missing smoking-history values, invalid numerical values, and incorrect data types have been handled correctly. Also verify that the newly created Age Group column contains the categories Young, Middle Age, and Senior.

15. To verify the ETL process visually, switch to Report view and create a Clustered Column Chart. Drag Age Group to the X-Axis field and drag blood\_glucose\_level to the Y-Axis field. Change the aggregation of blood\_glucose\_level to Average. This produces a visual comparison of the average blood glucose level among the different age groups.

**Average of diabetes**  
by gender



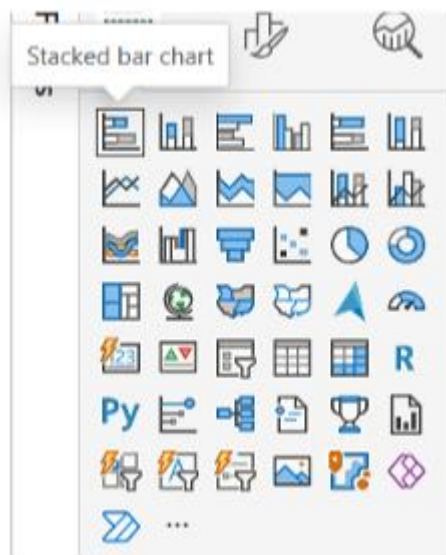
16. Verify that the chart displays the three standardized age groups: Young, Middle Age, and Senior. The resulting visual confirms that the diabetes data has been successfully extracted, transformed, and loaded into Power BI for further analysis.

### Interpretation

The clustered column chart compares the average blood glucose level across the three age groups. The age group with the tallest column has the highest average blood glucose level, while the age group with the shortest column has the lowest average blood glucose level.

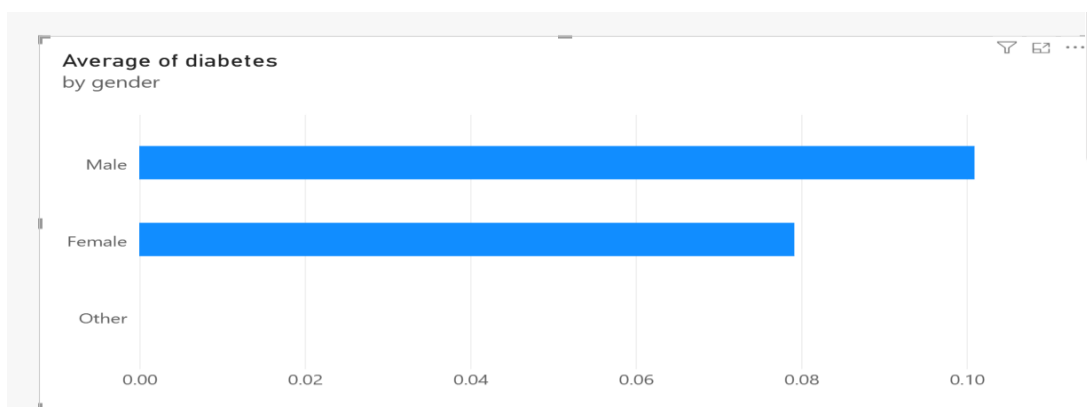
## Algorithm 3 — Creating a Bar Chart

### Step 1 Choose Bar Chart Visualization



### Step 2 Drag Age Group → Y-Axis

### Step 3 Drag blood\_glucose\_level → X-Axis



Step 4 Change blood\_glucose\_level aggregation from Sum to Average and format the chart.

Interpretation:

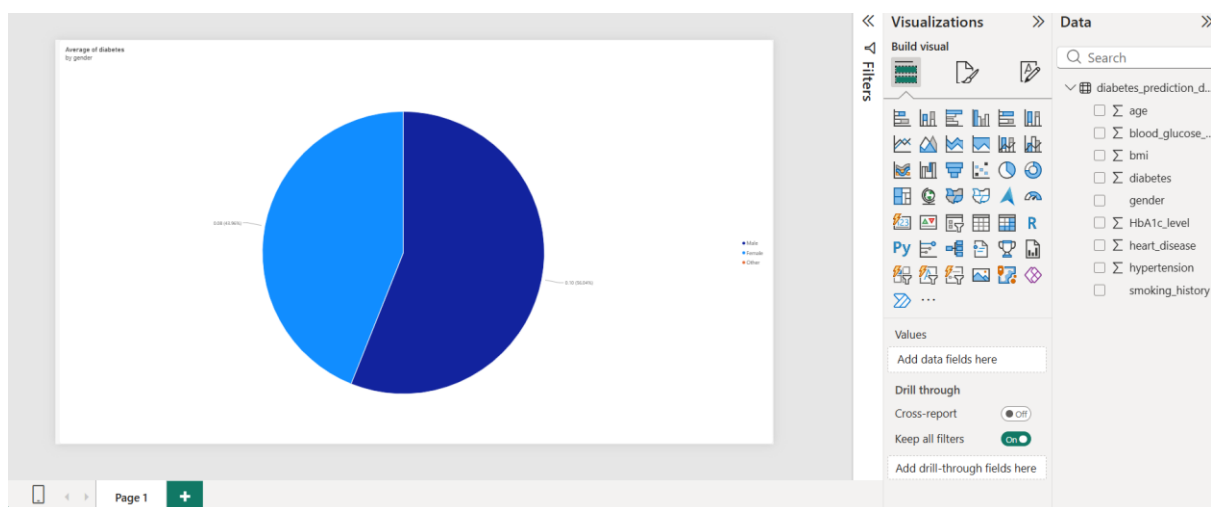
The age group with the highest bar generated the highest average blood glucose level, while the age group with the lowest bar generated the lowest average blood glucose level.

#### Algorithm 4 — Creating a Pie Chart

Step 1 Select Pie Chart

Step 2 Drag gender → Legend

Step 3 Drag diabetes → Values



Step 4 Set the aggregation of diabetes to Sum and format the chart.

**Output:**

A pie chart representing the distribution of diabetes cases between male and female patients.

**Interpretation:**

The gender category occupying the largest portion of the pie chart represents the highest number of diabetes cases.

#### Algorithm 5 — Creating a KPI Card

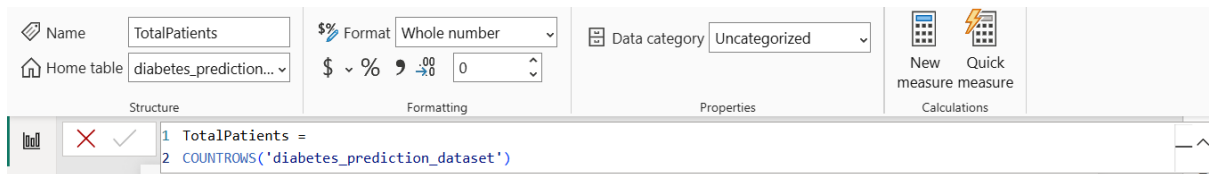
**Step 1** Create 4 measures by clicking on New Measure from the Measure Tools tab and type the formulas inside the DAX field:

**a. Total Patients**



TotalPatients =

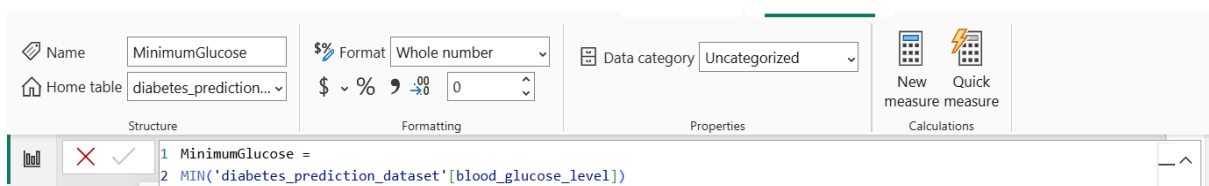
COUNTROWS('diabetes\_prediction\_dataset')



## b. Minimum Glucose

MinimumGlucose =

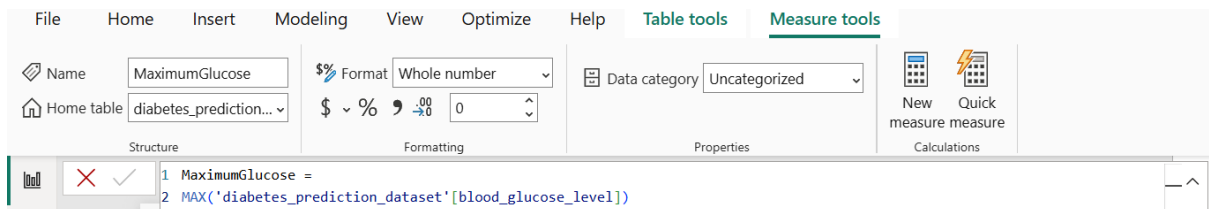
MIN('diabetes\_prediction\_dataset'[blood\_glucose\_level])



## c. Maximum Glucose

MaximumGlucose =

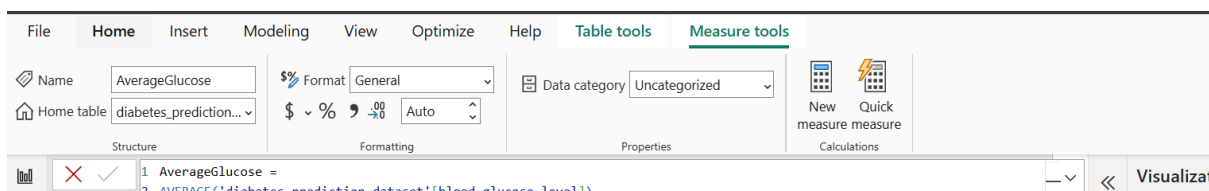
MAX('diabetes\_prediction\_dataset'[blood\_glucose\_level])



## d. Average Glucose

AverageGlucose =

AVERAGE('diabetes\_prediction\_dataset'[blood\_glucose\_level])

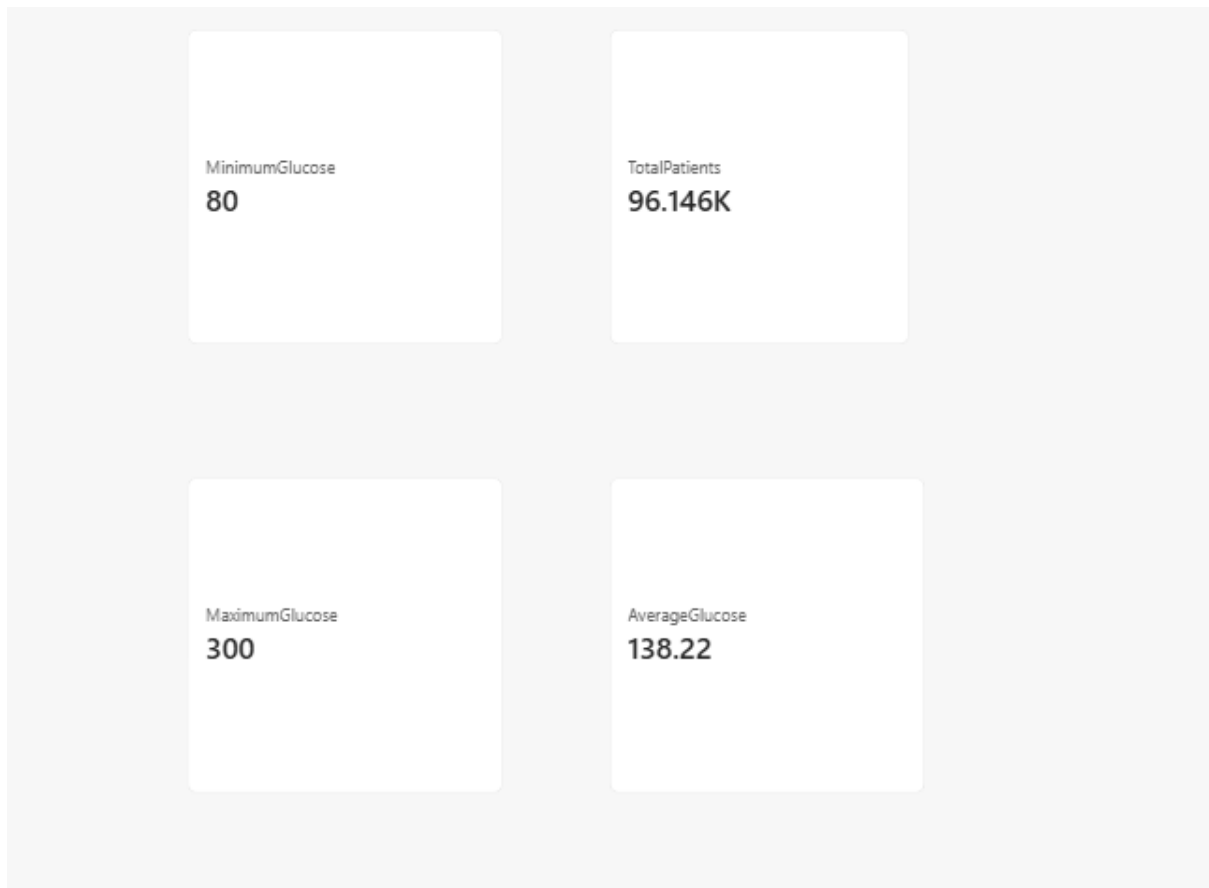


**Step 2 Create Card Visual.**

**Step 3 Drag the corresponding measure into the field.**

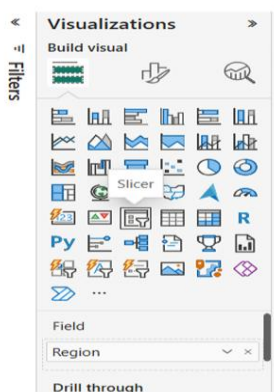
**Output:**

Four KPI cards displaying:



## Algorithm 6 — Creating Slicers

### Step 1 Select Slicer Visualization



Step 2 Drag smoking\_history Field

Step 3 Enable Single Selection

Step 4 Apply Filters

### Output:

Without any filter: Dashboard displays all patients.

With never filter: Dashboard displays patients with never smoking history.

With current filter: Dashboard displays patients with current smoking history.

With former filter: Dashboard displays patients with former smoking history.



## Algorithm 8 — Creating Dashboard

### Step 1 Add KPI Cards

- Total Patients
- Minimum Glucose
- Maximum Glucose
- Average Glucose

### Step 2 Add Bar Chart

- Age Group → Y-Axis
- Average Blood Glucose Level → X-Axis

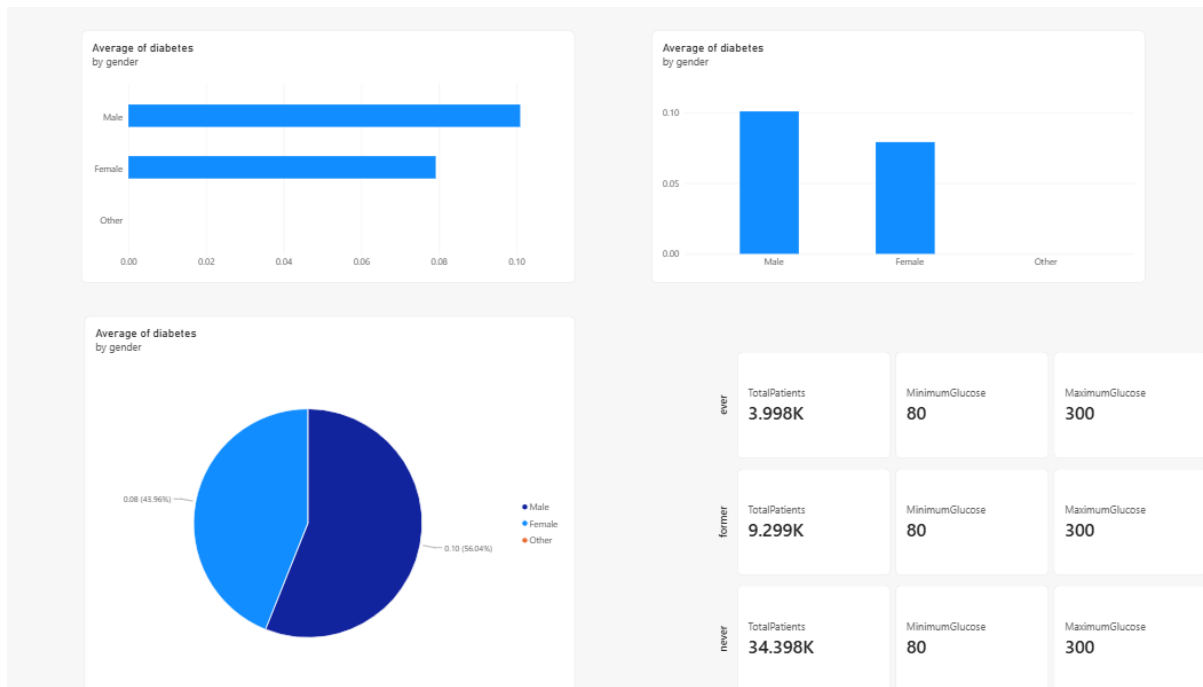
### Step 3 Add Pie Chart

- Gender → Legend
- Diabetes → Values

### Step 4 Add Slicer

- Age Group → Slicer

### Step 5 Arrange the visuals neatly on the dashboard.



**Step 6** Save the Power BI Dashboard.

### Output:

An interactive dashboard containing KPI cards, a bar chart, a pie chart, and an age-group slicer is created for analyzing the diabetes dataset.

### Result

The diabetes prediction data was successfully imported, transformed, analyzed, and visualized in Power BI Desktop, and an interactive dashboard was created for effective analysis and decision-making.